

EX NO:1 Butterworth IIR Filters

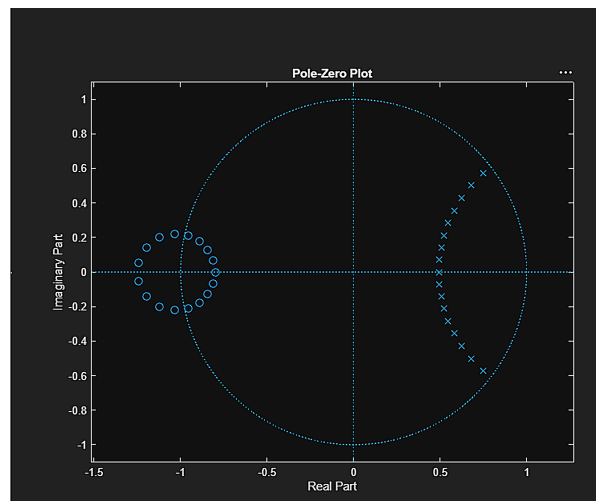
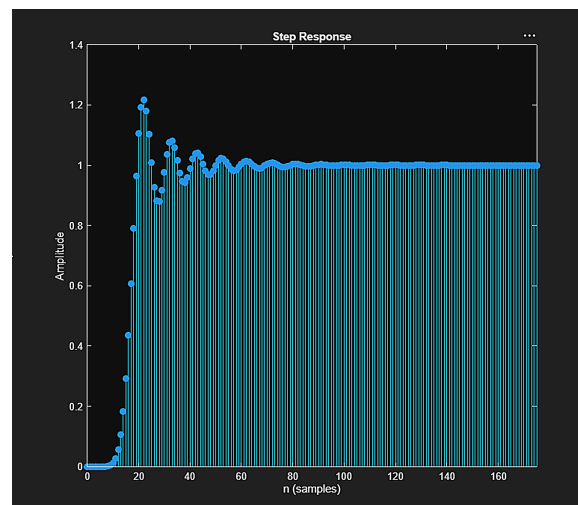
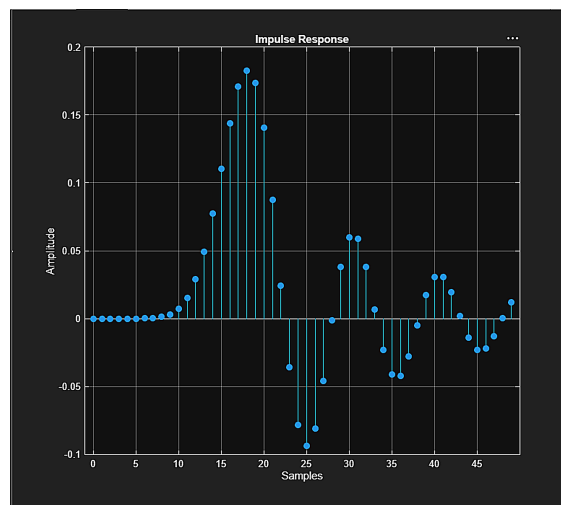
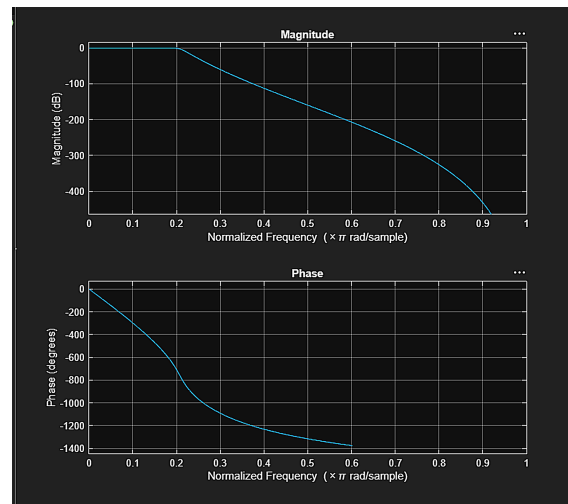
1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

```
1  clc;
2  clear;
3  close all;
4
5  fs = 1000;
6  Fs = 1000;
7  fst = 1500;
8
9
10 [N,wn] = buttord(fst/(Fs/2),fst/(Fs/2),1,40);
11 [b,a] = butter(N,wn);
12
13 % Frequency response
14 figure;
15 freqz(b,a);
16
17 % Impulse response
18 [h,s] = impz(b,a,50);
19 figure;
20 stem(h,'filled');
21 grid on;
22 xlabel('Samples');
23 ylabel('Amplitude');
24 title('Impulse Response');
25
26 % Step Response
27 figure;
28 step(b,a);
29
30 % Pole-Zero Plot
31 figure;
32 zplane(b,a);
33
34 fprintf('Minimum filter order (N) = %d\n',N);
35 fprintf('Cutoff frequency (wn) = %f\n',wn);
```

Command Window

New to MATLAB? See resources for [Getting Started](#).

Minimum Filter Order (N) = 17
Cutoff Frequency (wn) = 3.471e+003



2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

```
1 clear; close
2
3 f1=8000; f2=3000; f3=2000;
4 [b,a]=butter(10,(f1/2),f2/(f1/2),f3/(f1/2),1,'low');
5
6 % magnitude & phase
7 freqz(b,a)
8 % impulse response
9 figure
10 [h1,h2]=impz(b,a,50);
11 stem(h1), grid on
12 title('Impulse Response')
13 % step response
14 figure
15 step(b,a)
16 title('step response')
17 % group delay
18 figure
19 grpdelay(b,a)
20 title('group delay')
21 % pole-zero plot
22 figure
23 zplane(b,a)
24 title('Pole-zero plot')
25 fprintf('Minimum Filter Order (N) = %d\n',N);
26 fprintf('cutoff frequency (kHz) = %.4f\n',f1/1000);
27
28 Command Window
29
30 New to MATLAB? See resources for Getting Started.
31
32 Minimum Filter Order (N) = 12
33 cutoff frequency (kHz) = 8.5807
34
35 >
```

